Module 6 Data Cleaning Assignment

Magnificent Seven Financial Comparison Dashboard

Submission Information

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Branch	cleaning-nicole
Branch URL	https://github.com/xunyangy-afk/Python-Group-Project/tree/cleaning-nicole
Commit ID	fc5a3abbd98136955d74d5c295f6375887392a22 (current branch tip)
Data source	TheFinAI/Herculean
Source version or download date	Downloaded 2026-10-05; no upstream release identifier is recorded

Project Question

How can Herculean's financial data be transformed into an interactive dashboard for comparing the Magnificent Seven companies? The workflow prepares data for a reusable comparison of normalized stock performance while preserving the raw snapshot.

Row Reconciliation

Table	Source rows	ADBE excluded	Cleaned rows
Prices	2,656	332	2,324
News	3,888	486	3,402
Filings	73	10	63

The source snapshot includes ADBE; the cleaned outputs retain the seven target symbols only.

Assignment Checklist

1. STATE THE PROJECT VISION AND QUESTION | Done

The project vision is to build an interactive Magnificent Seven financial comparison dashboard using Herculean data. The business question is: how can the raw financial data be transformed into a reusable comparison tool for the seven major technology companies? The main metric is normalized stock performance for trend and comparison analysis.

2. DEFINE ONE ROW | Done

One price row is a single company-date observation; one news row is a single company-date story; one filing row is a single company-document-date filing record. The natural key for prices is (symbol, date). For news and filings, the natural keys are (symbol, date) and (symbol, date, document_type), respectively.

3. PRESERVE THE SOURCE | Done

Source data remain in `data/raw/` and are read only. The exact data source is TheFinAI/Herculean on Hugging Face.

4. INSPECT COLUMNS AND TYPES | Done

The raw prices table has 9 columns: id, symbol, date, open, high, low, close, adj_close, volume. News has 4 columns; filings has 6 columns. Date fields were parsed to datetime during cleaning, and all expected columns were inspected.

5. STANDARDIZE TEXT | Done

Symbol values were normalized to uppercase and whitespace was trimmed. No identifier values were altered beyond this formatting standardization; all other raw text is retained.

6. INVESTIGATE DUPLICATES | Done

Raw datasets had zero exact duplicate rows, zero repeated IDs, and zero repeated symbol-date combinations. No deduplication was required beyond the project-scope filter.

7. HANDLE MISSING DATA | Done

There were zero missing values in raw tables and the filtered M7 dataset. No imputation was required; missing values were retained as missing only if encountered in future data, with no fabricated replacement.

8. CHECK SUSPICIOUS VALUES | Done

The price validation checks resulted in zero negative prices, zero negative volume rows, zero case of high < low, and zero open/close values outside the daily range. News and filings were checked for blank text and invalid filing types; no invalid values were found.

9. VERIFY MERGES | Not applicable

The workflow does not merge the three tables for the cleaning step because each table is cleaned and analyzed independently before downstream dashboard assembly. If a later dashboard step merges tables, the merge keys are symbol and date.

10. RECONCILE CHANGES | Done

Starting totals were 2656 price rows, 3888 news rows, and 73 filing rows. After removing ADBE, final totals were 2324 price rows, 3402 news rows, and 63 filing rows. The differences are exactly the Adobe rows removed for scope.

11. CHECK CALCULATIONS | Done

Daily return is computed as close.pct_change(); normalized price is close / first close in each symbol series; rolling volume median is the 30-day median; volatility is the 30-day standard deviation of daily return; period is year-month; missing values in daily return or feature calculations remain as missing or zero only when a formula requires it. No fabricated values were introduced.

12. VERIFY INDIVIDUAL RECORDS | Done

The five record checks are included in the PDF and evidence summary; they cover difficult values and validation decisions.

13. TEST REPRODUCIBILITY | Done

The script was run from the original raw data and regenerated the same outputs in a second execution. The outputs were checked for row counts and hashes, and they matched.

14. DOCUMENT REMAINING LIMITATIONS | Done

The project data are valid and internally consistent for the M7 scope. The remaining limitation is that the assignment is based on the static raw source snapshot rather than a live market feed; additional data and a broader time horizon would be needed for future live dashboard updates.

Cleaning Decisions

Rule	Decision and reason	Verification
1. Project-scope filter to Magnificent Seven symbols	Keep only AAPL, MSFT, GOOGL, AMZN, NVDA, META, and TSLA. Exclude ADBE. The project objective is to compare the Magnificent Seven companies, and Adobe is outside the approved list.	Count rows per symbol before and after filtering; confirm ADBE is removed and M7 counts are consistent across all tables.
2. Standardize symbols and preserve case semantics	Strip whitespace and uppercase all symbol values before analysis. This prevents case-variant duplicates and ensures the same company is treated consistently.	Ensure the cleaned symbol set contains only the seven permitted values.
3. Convert source dates to consistent datetime objects	Parse dates with pandas and preserve the timezone-naive date format used by the source. All time-based calculations require valid dates for return, month, and time series analysis.	Confirm dates parse without invalid entries and that date ranges remain consistent with the source.
4. Validate pricing integrity before analysis	Reject any rows with negative prices or volume, and require high >= low plus open/close within the daily range. Any violations raise an error for review instead of silently modifying values. These checks protect the project from impossible price records and ensure the dashboard is built on valid market data.	Run validation after filtering and report zero invalid rows in the cleaned data.
5. Preserve uncertain text without invented replacements	Keep the original text exactly as provided unless it is blank, which is a flag for investigation rather than replacement. Do not invent text. The project dashboard must preserve the original reporting context and uncertainty without fabricating records.	Verify no blank strings are introduced and no synthetic text replaces source material.
6. Restrict filing records to 10-K and 10-Q	Keep only 10-K and 10-Q entries and discard all others if present. The project scope is limited to annual and quarterly SEC filings.	Confirm the cleaned filing data contains only 10-K and 10-Q values.
7. Add analysis features only from validated data	Compute fields directly from valid cleaned data and document them clearly. These features are needed for comparison and dashboard analysis but must remain transparent and reproducible.	Run the workflow twice and confirm identical computed columns and row counts.

Five Record Checks

Check 1: Price range integrity	
Source value	symbol=AAPL; date=2024-12-02; open=237.27000427246097; high=240.7899932861328; low=237.16000366210935; close=239.58999633789065
Expected	high >= low and open/close within the trading range
Actual	high=240.7899932861328, low=237.16000366210935, open=237.27000427246097, close=239.58999633789065
Result	PASS

Check 2: Volume positivity	
Source value	symbol=NVDA; date=2025-01-27; volume=818830900
Expected	volume should be positive and greater than zero
Actual	818830900
Result	PASS

Check 3: News text preservation	
Source value	symbol=AAPL; date=2024-12-01; highlights=Apple’s December 1, 2024 coverage centers on its fiscal first quarter 2024 results, with Shacknews reporting that the company generated revenue of $119.6 billion and earned $2.18 per share, figures that surpassed the consensus estimates of $118 billion in revenue and $2.10 in EPS. This framing positions the quarter as a positive earnings execution relative to expectations, highlighting a solid top-line and bottom-line performance. The article additionally draws market context from Statmuse, which is cited for stock-market data and related context around Apple on that date. This sourcing implies a broader market narrative surrounding Apple, incorporating how the stock had been moving and how the year-to-date trajectory was interacting with the earnings release, though the piece does not provide explicit price figures or forward-looking guidance within its summary. Regarding sentiment, the article itself does not assign an official sentiment score (noted as N/A in the source). Nevertheless, the reported earnings beat on both revenue and EPS is presented in a manner that implies a favorable view of Apple’s quarterly execution, at least in the immediate aftermath of the results. Key themes that emerge from the article include the emphasis on quarterly earnings strength and the integration of multiple data sources to frame Apple’s performance within the broader market context. The developments highlight how Apple’s Q1 results are being communicated alongside market data to give readers a snapshot of both financial performance and investor awareness on that date.
Expected	highlights text must be present and non-empty after stripping
Actual	Full highlights value preserved after surrounding whitespace normalization (1586 characters)
Result	PASS

Check 4: Filing type validity	
Source value	symbol=AAPL; date=2024-02-01; type=10-Q
Expected	document_type should be either 10-K or 10-Q
Actual	10-Q
Result	PASS

Check 5: ADBE scope exclusion	
Source value	symbol=ADBE; date=2024-12-02
Expected	ADBE must not appear in the cleaned M7 output
Actual	not present in cleaned M7 dataset
Result	PASS

Independent Codex Check

I used Codex to help implement and review the cleaning workflow. I independently checked Codex's recommendation to exclude Adobe by confirming that ADBE exists in the original Herculean tables but is not one of the Magnificent Seven. I compared the original and cleaned symbol lists and verified that ADBE was removed while all seven intended symbols remained. I reconciled the changes as follows: prices 2,656 – 332 = 2,324; news 3,888 – 486 = 3,402; and filings 73 – 10 = 63.